Activity_Course 3 TikTok project lab

October 28, 2025

1 TikTok Project

Course 3 - Go Beyond the Numbers: Translate Data into Insights

Your TikTok data team is still in the early stages of their latest project. So far, you've completed a project proposal and used Python to inspect and organize the TikTok dataset.

Orion Rainier, a Data Scientist at TikTok, is pleased with the work you have already completed and is requesting your assistance with some Exploratory Data Analysis (EDA) and data visualization. The management team asked to see a Python notebook showing data structuring and cleaning, as well as any matplotlib/seaborn visualizations plotted to help us understand the data. At the very least, include a graph comparing claim counts to opinion counts, as well as boxplots of the most important variables (like "video duration," "video like count," "video comment count," and "video view count") to check for outliers. Also, include a breakdown of "author ban status" counts.

Additionally, the management team has recently asked all EDA to include Tableau visualizations. Tableau visualizations are particularly helpful in status reports to the client and board members. For this data, create a Tableau dashboard showing a simple claims versus opinions count, as well as stacked bar charts of claims versus opinions for variables like video view counts, video like counts, video share counts, and video download counts. Make sure it is easy to understand to someone who isn't data savvy, and remember that the assistant director is a person with visual impairments.

You also notice a follow-up email from the Data Science Lead, Willow Jaffey. Willow suggests including an executive summary of your analysis to share with teammates.

A notebook was structured and prepared to help you in this project. Please complete the following questions.

2 Course 3 End-of-course project: Exploratory data analysis

In this activity, you will examine data provided and prepare it for analysis. You will also design a professional data visualization that tells a story, and will help data-driven decisions for business needs.

Please note that the Tableau visualization activity is optional, and will not affect your completion of the course. Completing the Tableau activity will help you practice planning out and plotting a data visualization based on a specific business need. The structure of this activity is designed to emulate the proposals you will likely be assigned in your career as a data professional. Completing this activity will help prepare you for those career moments.

The purpose of this project is to conduct exploratory data analysis on a provided data set. Your mission is to continue the investigation you began in C2 and perform further EDA on this data with the aim of learning more about the variables. Of particular interest is information related to what distinguishes claim videos from opinion videos.

The goal is to explore the dataset and create visualizations. *This activity has 4 parts:*

Part 1: Imports, links, and loading

Part 2: Data Exploration * Data cleaning

Part 3: Build visualizations

Part 4: Evaluate and share results

Follow the instructions and answer the question below to complete the activity. Then, you will complete an executive summary using the questions listed on the PACE Strategy Document.

Be sure to complete this activity before moving on. The next course item will provide you with a completed exemplar to compare to your own work.

3 Visualize a story in Tableau and Python

4 PACE stages

Throughout these project notebooks, you'll see references to the problem-solving framework PACE. The following notebook components are labeled with the respective PACE stage: Plan, Analyze, Construct, and Execute.

4.1 PACE: Plan

Consider the questions in your PACE Strategy Document and those below where applicable to craft your response: 1. Identify any outliers:

- What methods are best for identifying outliers?
- How do you make the decision to keep or exclude outliers from any future models?

To identify outliers, I will use statistical methods such as the interquartile range (IQR) and visual methods such as boxplots and histograms. Outliers are values that fall significantly outside the normal range of the data, typically below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

To decide whether to keep or remove them, I will consider the context of each variable. If an outlier results from data entry error or unrealistic value (for example, a video with billions of views), it will be excluded. However, if it represents an extreme but possible performance of a real TikTok video, it will be kept, as it may reflect genuine user engagement behavior

4.1.1 Task 1. Imports, links, and loading

Go to Tableau Public The following link will help you complete this activity. Keep Tableau Public open as you proceed to the next steps.

Link to supporting materials: Public Tableau: https://public.tableau.com/s/. Note that the TikTok dataset can be downloaded directly from this notebook by going to "Lab Files" in the menu bar at the top of the page, clicking into the "/home/jovyan/work" folder, selecting `tiktok_dataset.csv`, and clicking "Download" above the list of files.

For EDA of the data, import the packages that would be most helpful, such as `pandas`, `numpy`, `matplotlib.pyplot`, and `seaborn`.

[1]:
```
# Import packages for data manipulation
import pandas as pd
import numpy as np

# Import packages for data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

Then, load the dataset into a dataframe. Read in the data and store it as a dataframe object.

Note: As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

[2]:
```
# Load dataset into dataframe
data = pd.read_csv("tiktok_dataset.csv")
```

4.2 PACE: Analyze

Consider the questions in your PACE Strategy Document and those below where applicable to complete your code.

4.2.1 Task 2a: Data exploration and cleaning

The first step is to assess your data. Check the Data Source page on Tableau Public to get a sense of the size, shape and makeup of the data set.

Consider functions that help you understand and structure the data.

- `.head()`
- `.info()`
- `.describe()`
- `.groupby()`
- `.sort_values()`

Consider the following questions as you work:

What do you do about missing data (if any)?

Are there data outliers?

Start by discovering, using `.head()`, `.size`, and `.shape`.

[3]:
```
# Display and examine the first few rows of the dataframe
data.head()
```

[3]:
```
   # claim_status    video_id  video_duration_sec  \
0  1        claim  7017666017                  59
1  2        claim  4014381136                  32
2  3        claim  9859838091                  31
3  4        claim  1866847991                  25
4  5        claim  7105231098                  19

                           video_transcription_text verified_status  \
0  someone shared with me that drone deliveries a…    not verified
1  someone shared with me that there are more mic…    not verified
2  someone shared with me that american industria…    not verified
3  someone shared with me that the metro of st. p…    not verified
4  someone shared with me that the number of busi…    not verified

  author_ban_status  video_view_count  video_like_count  video_share_count  \
0      under review          343296.0           19425.0              241.0
1            active          140877.0           77355.0            19034.0
2            active          902185.0           97690.0             2858.0
3            active          437506.0          239954.0            34812.0
4            active           56167.0           34987.0             4110.0

   video_download_count  video_comment_count
0                   1.0                  0.0
1                1161.0                684.0
2                 833.0                329.0
3                1234.0                584.0
4                 547.0                152.0
```

[4]:
```
# Get the size of the data
data.size
```

[4]:
```
232584
```

[5]:
```
# Get the shape of the data
data.shape
```

[5]:
```
(19382, 12)
```

Get basic information about the data, using `.info()`.

[6]:
```python
# Get basic information about the data
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19382 entries, 0 to 19381
Data columns (total 12 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   #                         19382 non-null  int64
 1   claim_status              19084 non-null  object
 2   video_id                  19382 non-null  int64
 3   video_duration_sec        19382 non-null  int64
 4   video_transcription_text  19084 non-null  object
 5   verified_status           19382 non-null  object
 6   author_ban_status         19382 non-null  object
 7   video_view_count          19084 non-null  float64
 8   video_like_count          19084 non-null  float64
 9   video_share_count         19084 non-null  float64
 10  video_download_count      19084 non-null  float64
 11  video_comment_count       19084 non-null  float64
dtypes: float64(5), int64(3), object(4)
memory usage: 1.8+ MB
```

Generate a table of descriptive statistics, using `.describe()`.

[7]:
```python
# Generate a table of descriptive statistics
data.describe()
```

[7]:
```
                  #      video_id  video_duration_sec  video_view_count  \
count  19382.000000  1.938200e+04        19382.000000      19084.000000
mean    9691.500000  5.627454e+09           32.421732     254708.558688
std     5595.245794  2.536440e+09           16.229967     322893.280814
min        1.000000  1.234959e+09            5.000000         20.000000
25%     4846.250000  3.430417e+09           18.000000       4942.500000
50%     9691.500000  5.618664e+09           32.000000       9954.500000
75%    14536.750000  7.843960e+09           47.000000     504327.000000
max    19382.000000  9.999873e+09           60.000000     999817.000000

       video_like_count  video_share_count  video_download_count  \
count      19084.000000       19084.000000          19084.000000
mean       84304.636030       16735.248323           1049.429627
std       133420.546814       32036.174350           2004.299894
min            0.000000           0.000000              0.000000
25%          810.750000         115.000000              7.000000
50%         3403.500000         717.000000             46.000000
75%       125020.000000       18222.000000           1156.250000
max       657830.000000      256130.000000          14994.000000
```

```
       video_comment_count
count         19084.000000
mean            349.312146
std             799.638865
min               0.000000
25%               1.000000
50%               9.000000
75%             292.000000
max            9599.000000
```

4.2.2 Task 2b. Assess data types

In Tableau, staying on the data source page, double check the data types of the columns in the dataset. Refer to the dimensions and measures in Tableau.

Review the instructions linked in the previous Activity document to create the required Tableau visualization.

4.2.3 Task 2c. Select visualization type(s)

Select data visualization types that will help you understand and explain the data.

Now that you know which data columns you'll use, it is time to decide which data visualization makes the most sense for EDA of the TikTok dataset. What type of data visualization(s) would be most helpful? Consider the distribution of the data.

- Line graph
- Bar chart
- Box plot
- Histogram
- Heat map
- Scatter plot
- A geographic map

For the TikTok dataset, the most helpful visualizations for exploratory data analysis (EDA) are bar charts, box plots, histograms, scatter plots, and possibly a heat map.

Bar charts will compare the number of claim vs. opinion videos.

Box plots will help identify outliers and differences in engagement metrics (likes, views, comments) between the two categories.

Histograms will show the distribution of metrics such as video duration or view counts.

Scatter plots will explore relationships between numerical variables, such as views vs. likes.

Heat maps can display correlations between engagement variables.

Line graphs and geographic maps are not applicable because the dataset does not include time or location data.

4.3 PACE: Construct

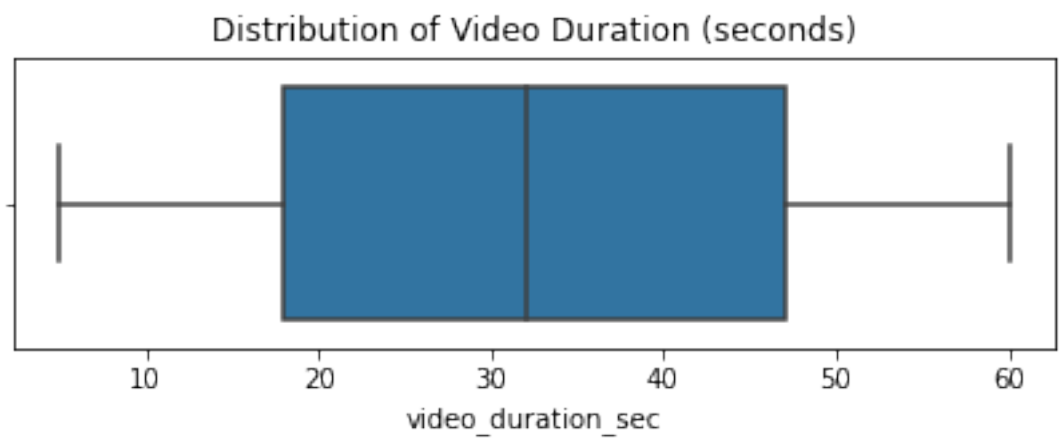
Consider the questions in your PACE Strategy Document to reflect on the Construct stage.

4.3.1 Task 3. Build visualizations

Now that you have assessed your data, it's time to plot your visualization(s).

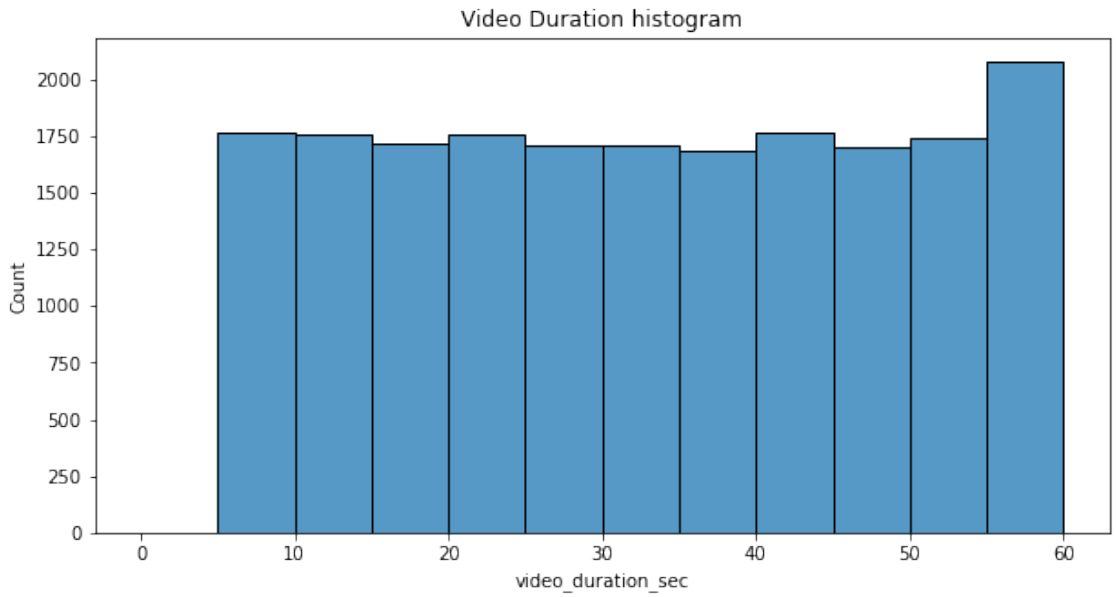
video_duration_sec Create a box plot to examine the spread of values in the `video_duration_sec` column.

[31]:
```
# Create a boxplot to visualize distribution of `video_duration_sec`
plt.figure(figsize=(7,2))
plt.title('Distribution of Video Duration (seconds)')
sns.boxplot(x=data['video_duration_sec'], fliersize=1);
```



Create a histogram of the values in the `video_duration_sec` column to further explore the distribution of this variable.

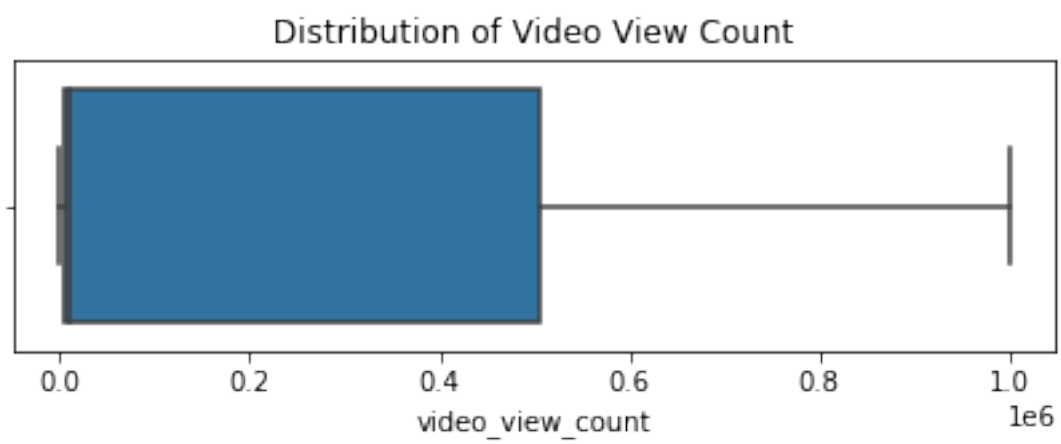
[9]:
```
# Create a histogram
plt.figure(figsize=(10,5))
sns.histplot(data['video_duration_sec'], bins=range(0,61,5))
plt.title('Video Duration histogram');
```



Question: What do you notice about the duration and distribution of the videos? All videos are 5-60 seconds in length, and the distribution is uniform.

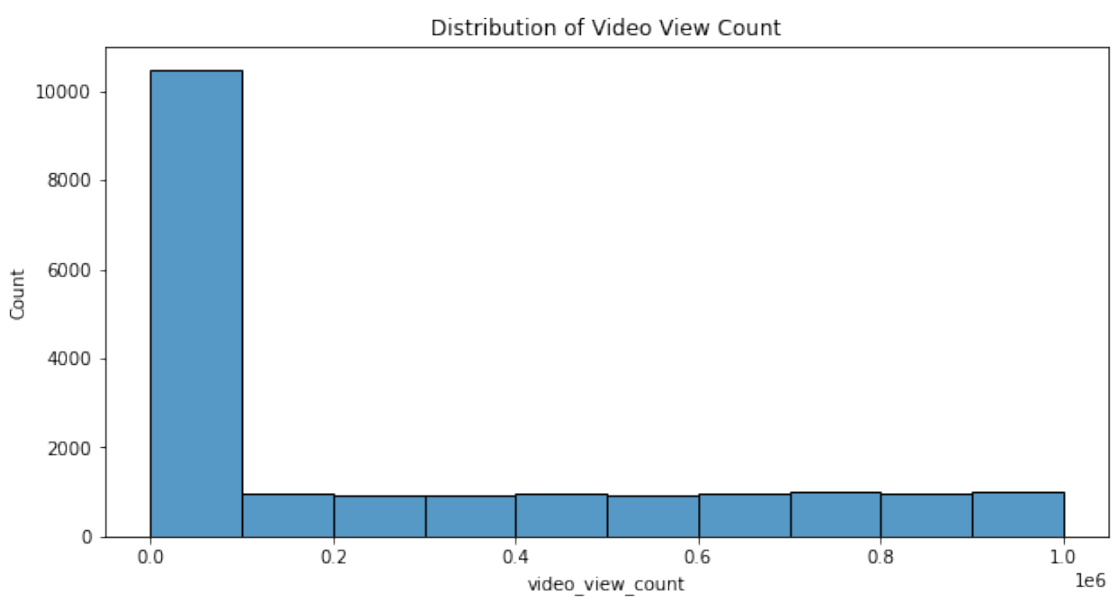
video_view_count Create a box plot to examine the spread of values in the `video_view_count` column.

[30]:
```
# Create a boxplot to visualize distribution of `video_view_count`
plt.figure(figsize=(7,2))
plt.title('Distribution of Video View Count')
sns.boxplot(x=data['video_view_count']);
```



Create a histogram of the values in the `video_view_count` column to further explore the distribution of this variable.

[15]:
```
# Create a histogram
"""
Note:
When creating histograms, the 'bins' parameter controls how data is grouped.

- Using bins=range(0, (10**6 + 1), 10**5):
  Defines fixed bin edges manually, creating intervals of exactly 100,000 units
  from 0 to 1,000,000. This approach ensures consistent bin sizes and alignment
  across multiple plots - ideal for comparing distributions (e.g., claim vs␣
 ↪opinion videos).

- Using bins=10:
  Automatically divides the full data range (from min to max) into 10 equal␣
 ↪parts.
  Bin widths adjust dynamically depending on the data range, which can slightly
  change between datasets.

In summary:
'bins=range(...)' → fixed, consistent intervals (manual control)
'bins=10' → automatic, data-dependent intervals (less precise but simpler)
"""
plt.figure(figsize=(10,5))
sns.histplot(data['video_view_count'], bins=range(0,(10**6+1),10**5))
plt.title('Distribution of Video View Count');
```

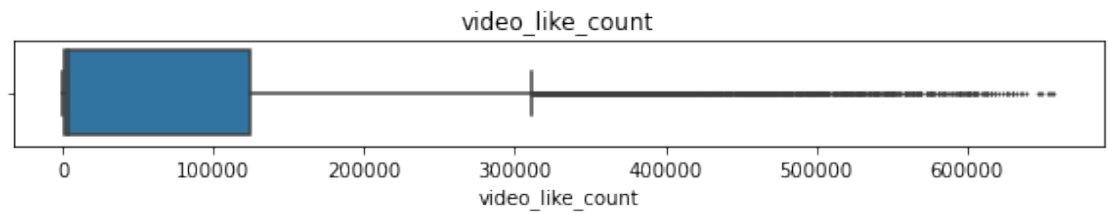


Question: What do you notice about the distribution of this variable?

This variable has a very uneven distribution, with more than half the videos receiving fewer than 100,000 views. Distribution of view counts > 100,000 views is uniform.

video_like_count Create a box plot to examine the spread of values in the `video_like_count` column.

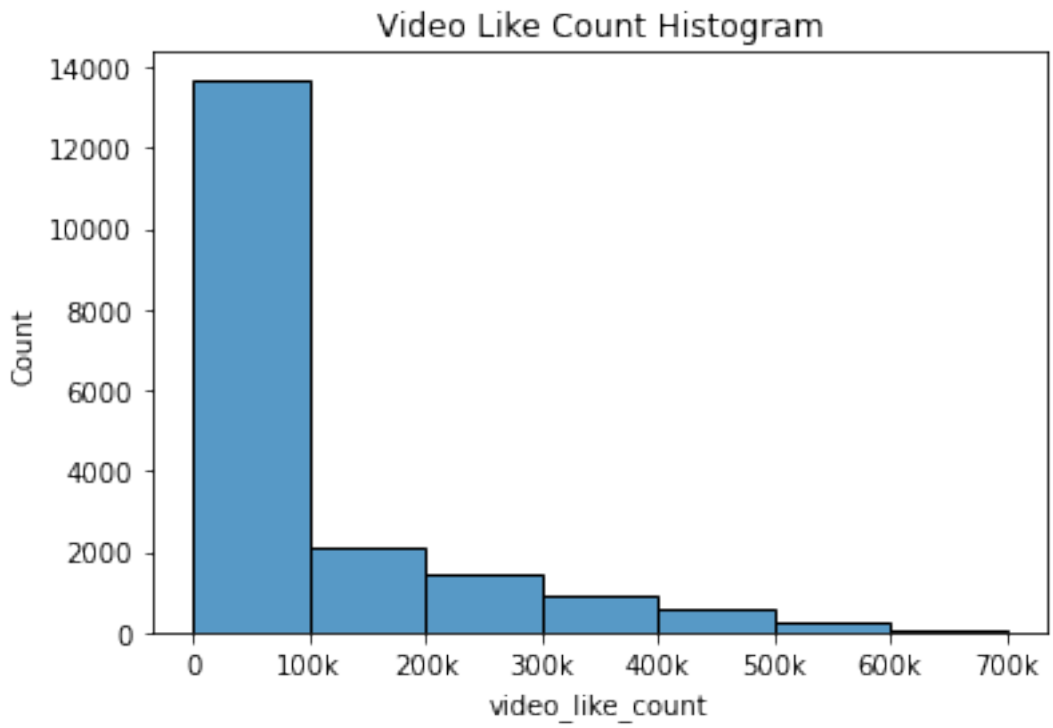
[29]:
```
# Create a boxplot to visualize distribution of `video_like_count`
# Keep outliers visible but minimal - fliersize=1 makes boxplots cleaner and␣
↪easier to read
plt.figure(figsize=(10,1))
plt.title('video_like_count')
sns.boxplot(x=data['video_like_count'], fliersize=1);
```



Create a histogram of the values in the `video_like_count` column to further explore the distribution of this variable.

[23]:
```
# Create a histogram

"""
Note:
Matplotlib no longer supports passing tick labels directly inside set_xticks().
Using a call such as:
    ax.set_xticks(ticks, labels=labels)
will raise:
    TypeError: got an unexpected keyword argument 'labels'

Explanation:
- The method set_xticks() now only accepts one argument: the tick positions.
- The tick labels must be set separately using set_xticklabels().

Correct usage:
    ax.set_xticks(range(0, 7*10**5 + 1, 10**5))   # Define tick positions
    ax.set_xticklabels(labels)                    # Define tick labels

Alternatively, you can use Matplotlib's FuncFormatter to automatically
format tick labels (e.g., converting 100000 to '100k') for cleaner plots:
```

```
    ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{int(x/
↪1000)}k'))

Summary:
set_xticks() → defines positions
set_xticklabels() → defines labels
"""

#plt.figure(figsize=(5,3))
ax = sns.histplot(data['video_like_count'], bins=range(0, (7*10**5 + 1), 10**5))
labels = [0] + [str(i) + 'k' for i in range(100, 701, 100)]
ax.set_xticks(range(0, 7*10**5 + 1, 10**5))
ax.set_xticklabels(labels)
plt.title('Video Like Count Histogram');
```

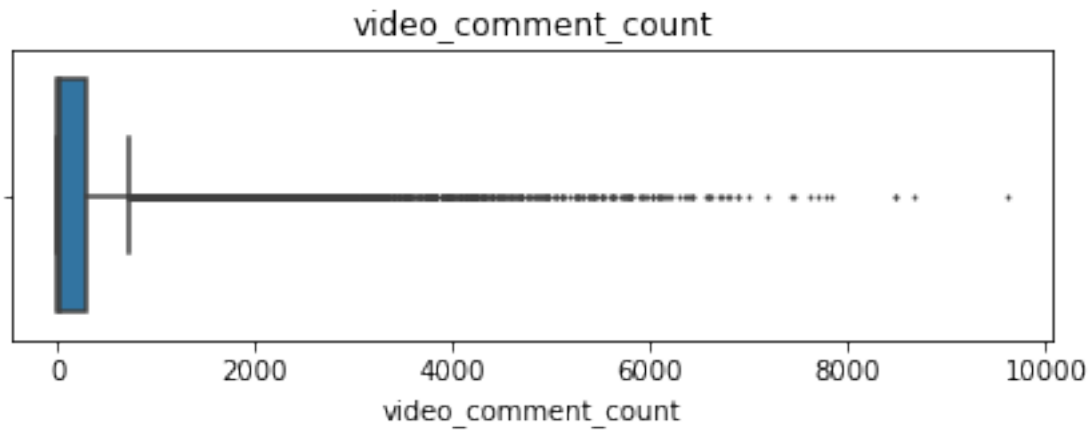


Question: What do you notice about the distribution of this variable?

Similar to view count, there are far more videos with $< 100{,}000$ likes than there are videos with more. However, in this case, there is more of a taper, as the data skews right, with many videos at the upper extremity of like count.

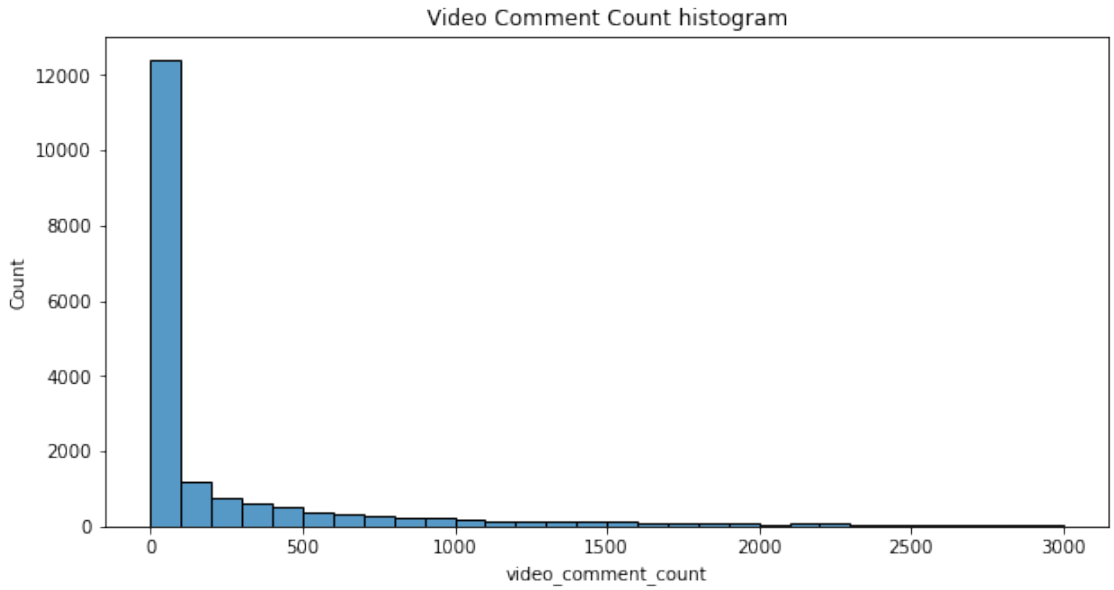
video_comment_count Create a box plot to examine the spread of values in the `video_comment_count` column.

[28]:
```
# Create a boxplot to visualize distribution of `video_comment_count`
# Keep outliers visible but minimal - fliersize=1 makes boxplots cleaner and␣
↪easier to read
plt.figure(figsize=(7,2))
plt.title('video_comment_count')
sns.boxplot(x=data['video_comment_count'], fliersize=1);
```



Create a histogram of the values in the `video_comment_count` column to further explore the distribution of this variable.

[26]:
```
# Create a histogram

"""
Note:
'bins=range(0, 3001, 100)' defines fixed histogram intervals from 0 to 3000,
in steps of 100 units. Each bin represents a range of 100 values
(e.g., 0-100, 100-200, ..., 2900-3000).

This manual bin range ensures consistent and evenly spaced intervals
for better comparison across different histograms.
"""
plt.figure(figsize=(10,5))
sns.histplot(data['video_comment_count'], bins=range(0,(3001),100))
plt.title('Video Comment Count histogram');
```

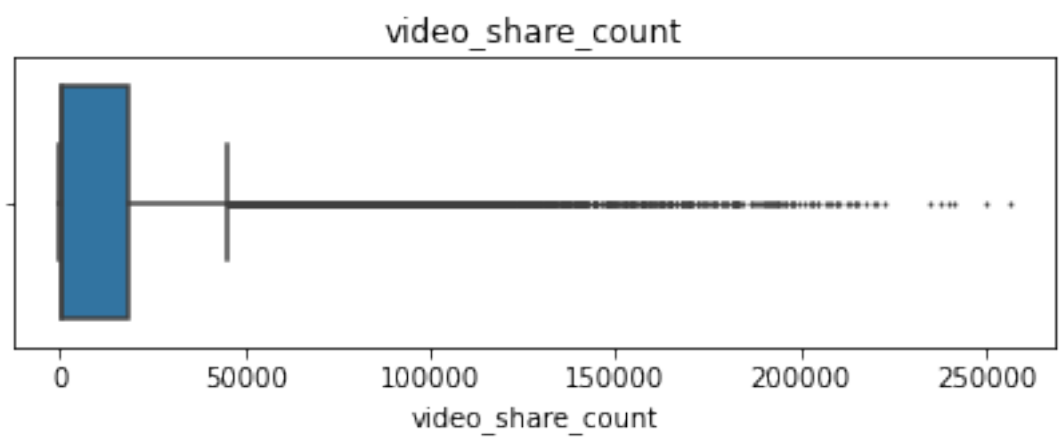


Question: What do you notice about the distribution of this variable?

Again, the vast majority of videos are grouped at the bottom of the range of values for video comment count. Most videos have fewer than 100 comments. The distribution is very right-skewed.

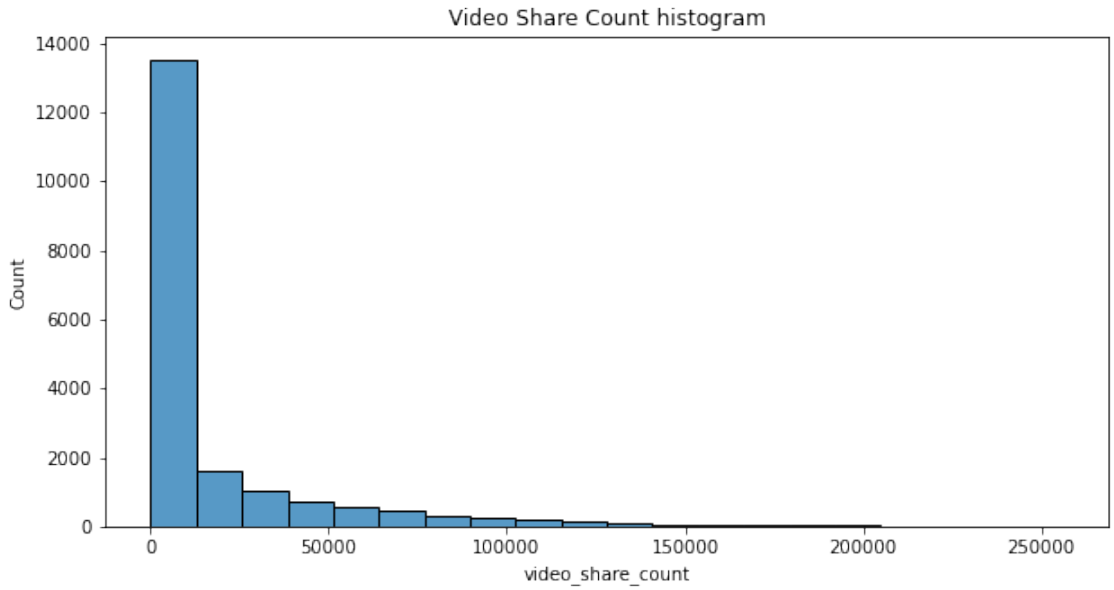
video_share_count Create a box plot to examine the spread of values in the `video_share_count` column.

[27]:
```
# Create a boxplot to visualize distribution of `video_share_count`
plt.figure(figsize=(7,2))
plt.title('video_share_count')
sns.boxplot(x=data['video_share_count'], fliersize=1);
```



Create a histogram of the values in the `video_share_count` column to further explore the distribution of this variable.

[17]:
```
# Create a histogram
plt.figure(figsize=(10,5))
sns.histplot(data['video_share_count'], bins=20)
plt.title('Video Share Count histogram');
```

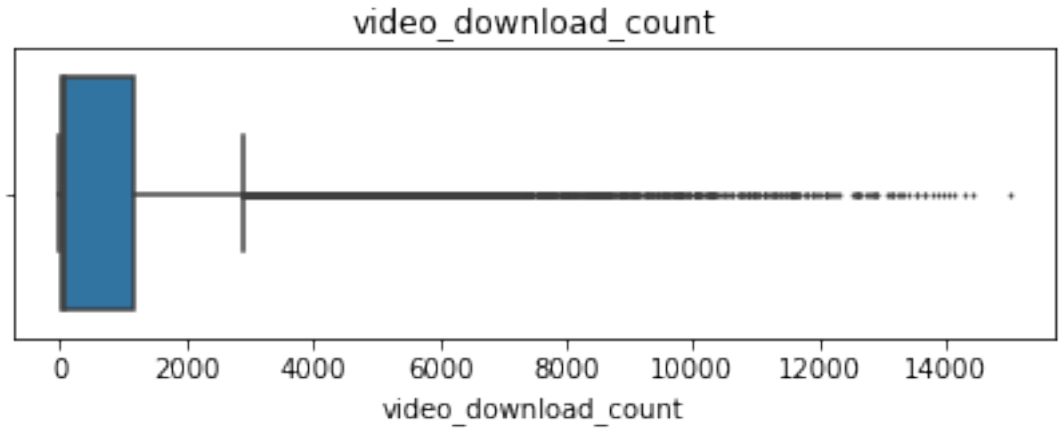


Question: What do you notice about the distribution of this variable?

The overwhelming majority of videos had fewer than 10,000 shares. The distribution is very skewed to the right.

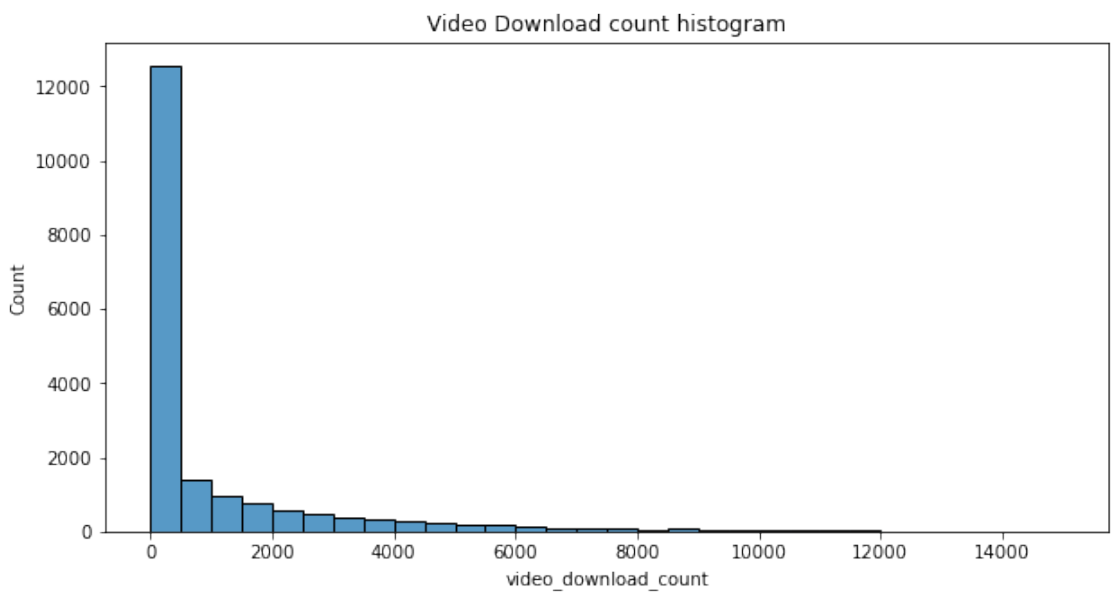
video_download_count Create a box plot to examine the spread of values in the `video_download_count` column.

[32]:
```
# Create a boxplot to visualize distribution of `video_download_count`
plt.figure(figsize=(7,2))
plt.title('video_download_count')
sns.boxplot(x=data['video_download_count'], fliersize=1);
```



Create a histogram of the values in the `video_download_count` column to further explore the distribution of this variable.

[34]:
```
# Create a histogram
plt.figure(figsize=(10,5))
sns.histplot(data['video_download_count'], bins=range(0,(15001),500))
plt.title('Video Download count histogram');
```

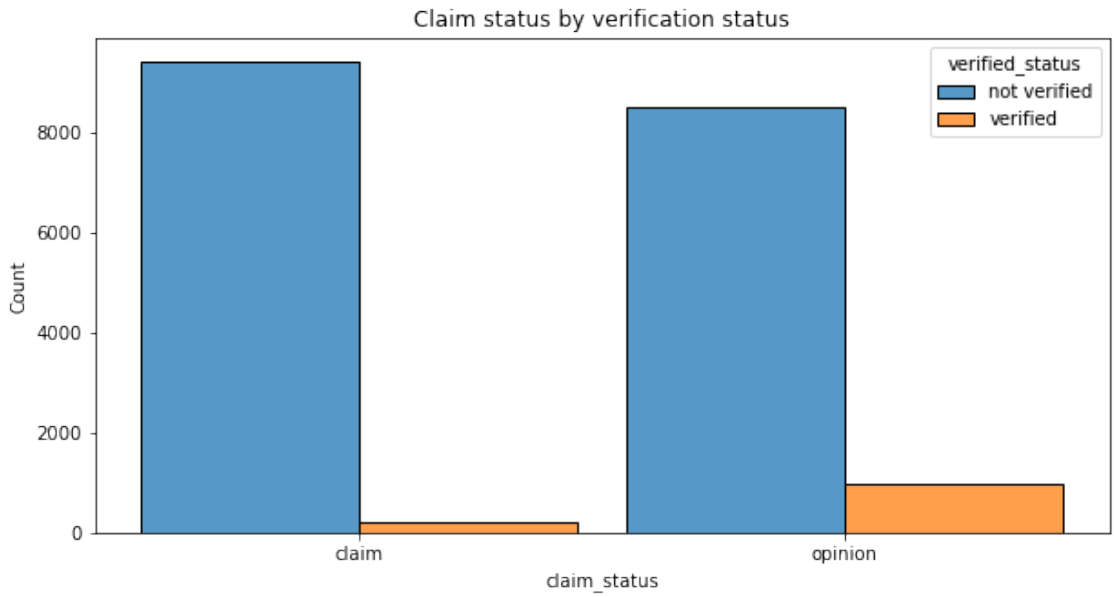


Question: What do you notice about the distribution of this variable?

The majority of videos were downloaded fewer than 500 times, but some were downloaded over 12,000 times. Again, the data is very skewed to the right.

Claim status by verification status Now, create a histogram with four bars: one for each combination of claim status and verification status.

[42]:
```
# Create a histogram

"""
Note:
The parameters 'multiple' and 'shrink' control how grouped bars are displayed
when using the 'hue' argument in Seaborn's histplot().

- multiple:
  Determines how histograms for different hue categories are drawn.
    'layer'  → bars overlap with transparency (default)
    'stack'  → bars are stacked on top of each other
    'fill'   → bars are stacked and normalized to 100% height
    'dodge'  → bars are placed side by side for direct comparison
  In this plot, 'multiple="dodge"' is used to compare 'verified' vs 'unverified'
  users across each 'claim_status' category.

- shrink:
  Adjusts the width of the bars (value between 0 and 1).
    shrink=1.0 → full width (bars touch)
    shrink=0.9 → slightly narrower bars, leaving small gaps
    shrink=0.5 → much thinner bars
  Here, 'shrink=0.9' improves readability by adding a small space
  between side-by-side bars.

Summary:
'multiple="dodge"' → displays bars side by side
'shrink=0.9' → makes bars slightly narrower for cleaner visualization
"""

plt.figure(figsize=(10,5))
sns.histplot(data=data, x='claim_status', hue='verified_status',␣
 ↪multiple='dodge', shrink=0.9)
plt.title('Claim status by verification status');
```



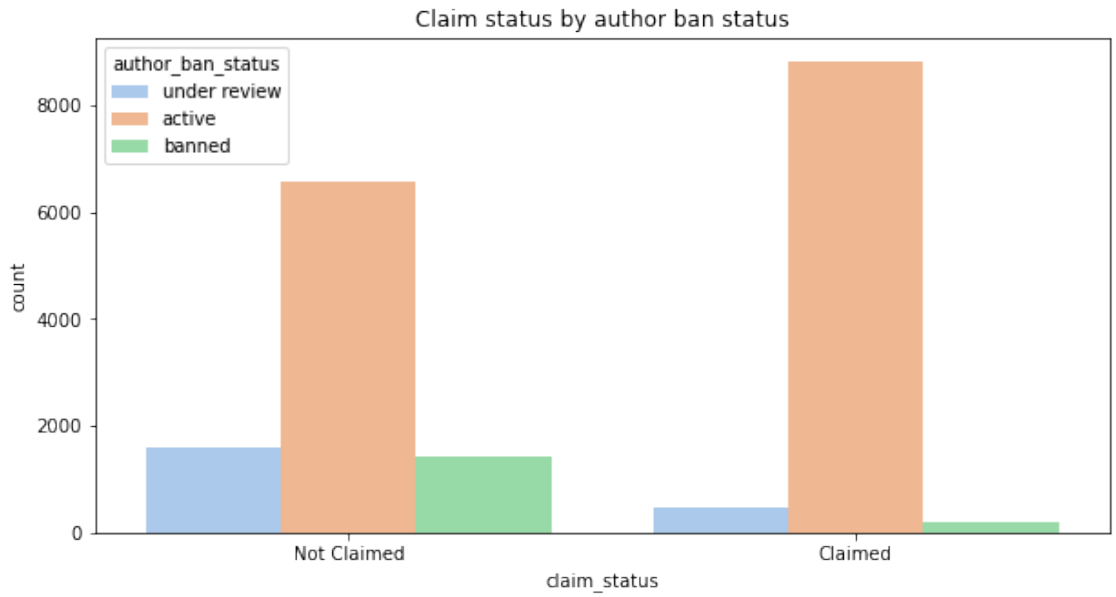
Question: What do you notice about the number of verified users compared to unverified? And how does that affect their likelihood to post opinions?

There are far fewer verified users than unverified users, but if a user is verified, they are much more likely to post opinions.

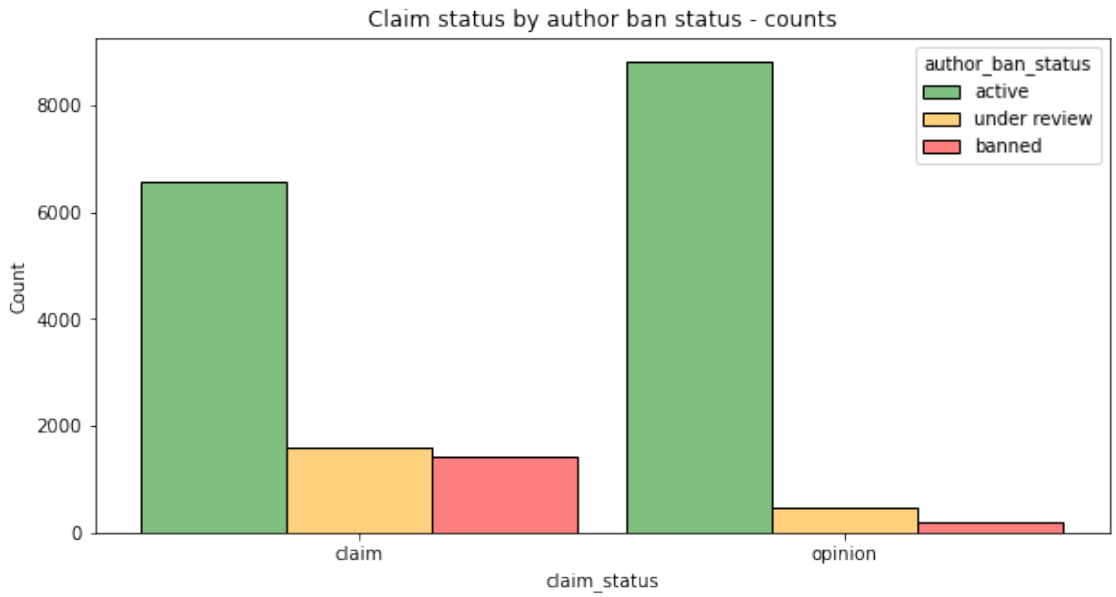
Claim status by author ban status The previous course used a `groupby()` statement to examine the count of each claim status for each author ban status. Now, use a histogram to communicate the same information.

[21]:
```
# Create a histogram

"""
Note:
Use sns.countplot() when you want to visualize the frequency of categorical␣
↪data.
This function automatically counts the number of occurrences for each category
on the x-axis and separates them by hue (if provided).

It is best suited for:
- Comparing category counts (e.g., "Claimed" vs. "Not Claimed")
- Displaying discrete data distributions
- Quick frequency checks without numerical bins

For more control over bar width, transparency, and spacing,
use sns.histplot() instead, which provides additional styling flexibility.
"""
```

```
plt.figure(figsize=(10,5))
sns.countplot(data=data, x='claim_status', hue='author_ban_status',␣
↪palette='pastel')
plt.xticks([0, 1], ['Not Claimed', 'Claimed'])
plt.title('Claim status by author ban status');
```



[45]:
```
fig =plt.figure(figsize=(10,5))
sns.histplot(data, x='claim_status', hue='author_ban_status',
             multiple='dodge',
             hue_order=['active', 'under review', 'banned'],
             shrink=0.9,
             palette={'active':'green', 'under review': 'orange', 'banned':␣
↪'red'},
             alpha=0.5)
plt.title('Claim status by author ban status - counts');
```

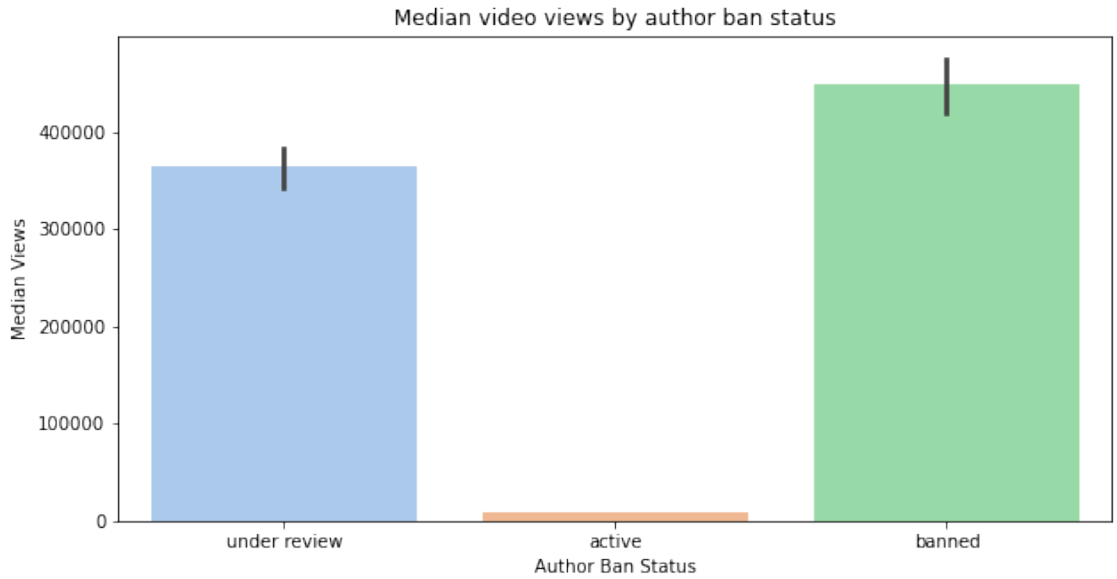


Question: What do you notice about the number of active authors compared to banned authors for both claims and opinions?

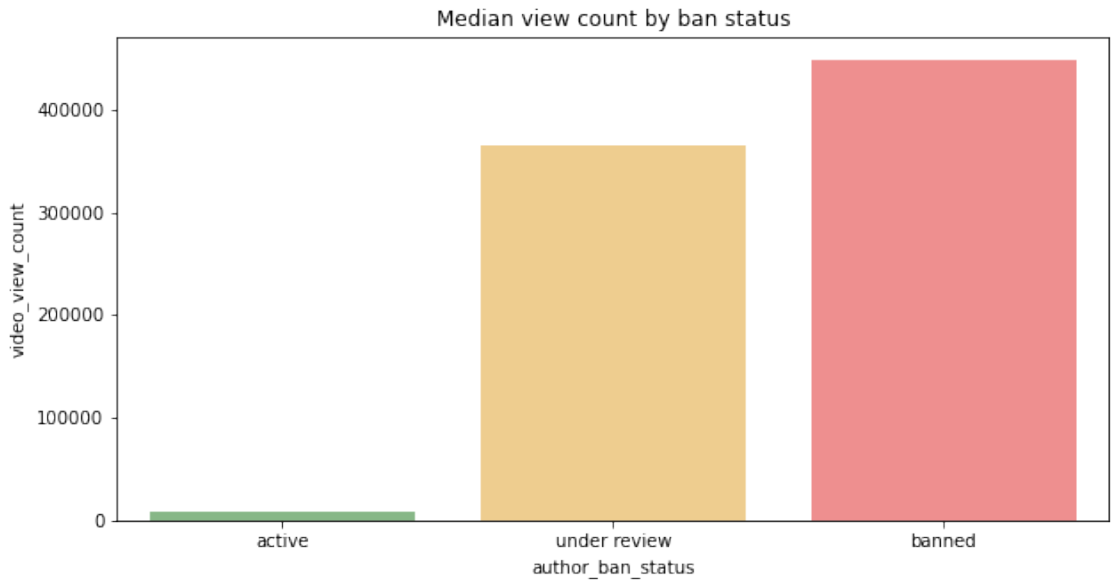
For both claims and opinions, there are many more active authors than banned authors or authors under review; however, the proportion of active authors is far greater for opinion videos than for claim videos. Again, it seems that authors who post claim videos are more likely to come under review and/or get banned.

Median view counts by ban status Create a bar plot with three bars: one for each author ban status. The height of each bar should correspond with the median number of views for all videos with that author ban status.

[22]:
```
# Create a bar plot
plt.figure(figsize=(10,5))
sns.barplot(data=data, x='author_ban_status', y='video_view_count',␣
 ↪estimator=np.median, palette='pastel')
plt.title('Median video views by author ban status')
plt.ylabel('Median Views')
plt.xlabel('Author Ban Status');
```



[48]:
```
# Create a bar plot Exemplar
ban_status_counts = data.groupby(['author_ban_status']).median(
    numeric_only=True).reset_index()

fig =plt.figure(figsize=(10,5))
sns.barplot(data=ban_status_counts,
           x='author_ban_status',
           y='video_view_count',
           order=['active', 'under review', 'banned'],
           palette={'active': 'green', 'under review':'orange', 'banned':␣
 ↪'red'},
           alpha=0.5)
plt.title('Median view count by ban status');
```



Question: What do you notice about the median view counts for non-active authors compared to that of active authors? Based on that insight, what variable might be a good indicator of claim status?

The median view counts for non-active authors are many times greater than the median view count for active authors. Since you know that non-active authors are more likely to post claims, and that videos by non-active authors get far more views on aggregate than videos by active authors, then `video_view_count` might be a good indicator of claim status.

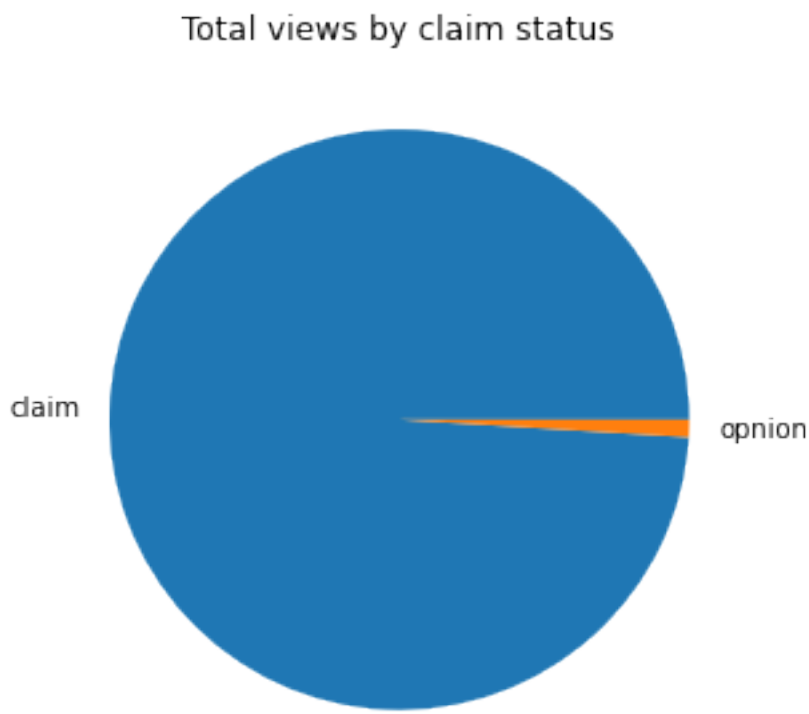
Indeed, a quick check of the median view count by claim status bears out this assessment:

[49]:
```
# Calculate the median view count for claim status.
data.groupby('claim_status')['video_view_count'].median()
```

[49]:
```
claim_status
claim      501555.0
opinion      4953.0
Name: video_view_count, dtype: float64
```

Total views by claim status Create a pie graph that depicts the proportions of total views for claim videos and total views for opinion videos.

[50]:
```
# Create a pie graph
fig = plt.figure(figsize=(10,5))
plt.pie(data.groupby('claim_status')['video_view_count'].sum(),␣
 ↪labels=['claim', 'opnion'])
plt.title('Total views by claim status');
```

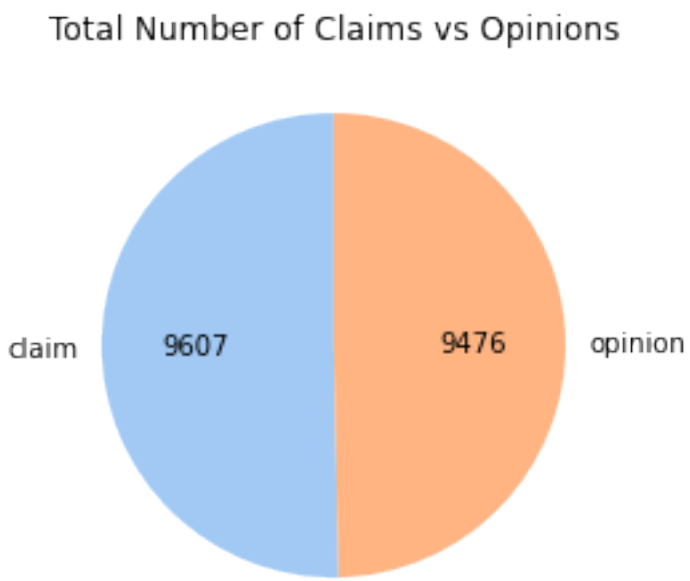


Question: What do you notice about the overall view count for claim status?

The overall view count is dominated by claim videos even though there are roughly the same number of each video in the dataset.

To validate the Tableau results, I generated the same pie chart in Python and confirmed the total counts for claims and opinions.

[25]:
```
# Count how many times 'claim' and 'opinion' appear
counts = data['claim_status'].value_counts()

# Create pie chart with pastel colors
colors = sns.color_palette('pastel')

# Creates pie chart with absolute values
plt.pie(counts, labels=counts.index, autopct=lambda p: int(p * sum(counts) /␣
↪100), startangle=90, colors=colors)
plt.title('Total Number of Claims vs Opinions');
```



4.3.2 Task 4. Determine outliers

When building predictive models, the presence of outliers can be problematic. For example, if you were trying to predict the view count of a particular video, videos with extremely high view counts might introduce bias to a model. Also, some outliers might indicate problems with how data was captured or recorded.

The ultimate objective of the TikTok project is to build a model that predicts whether a video is a claim or opinion. The analysis you've performed indicates that a video's engagement level is strongly correlated with its claim status. There's no reason to believe that any of the values in the TikTok data are erroneously captured, and they align with expectation of how social media works: a very small proportion of videos get super high engagement levels. That's the nature of viral content.

Nonetheless, it's good practice to get a sense of just how many of your data points could be considered outliers. The definition of an outlier can change based on the details of your project, and it helps to have domain expertise to decide a threshold. You've learned that a common way to determine outliers in a normal distribution is to calculate the interquartile range (IQR) and set a threshold that is 1.5 * IQR above the 3rd quartile.

In this TikTok dataset, the values for the count variables are not normally distributed. They are heavily skewed to the right. One way of modifying the outlier threshold is by calculating the **median** value for each variable and then adding 1.5 * IQR. This results in a threshold that is, in this case, much lower than it would be if you used the 3rd quartile.

Write a for loop that iterates over the column names of each count variable. For each iteration: 1. Calculate the IQR of the column 2. Calculate the median of the column 3. Calculate the outlier

threshold (median + 1.5 * IQR) 4. Calculate the numer of videos with a count in that column that exceeds the outlier threshold 5. Print "Number of outliers, {column name}: {outlier count}"

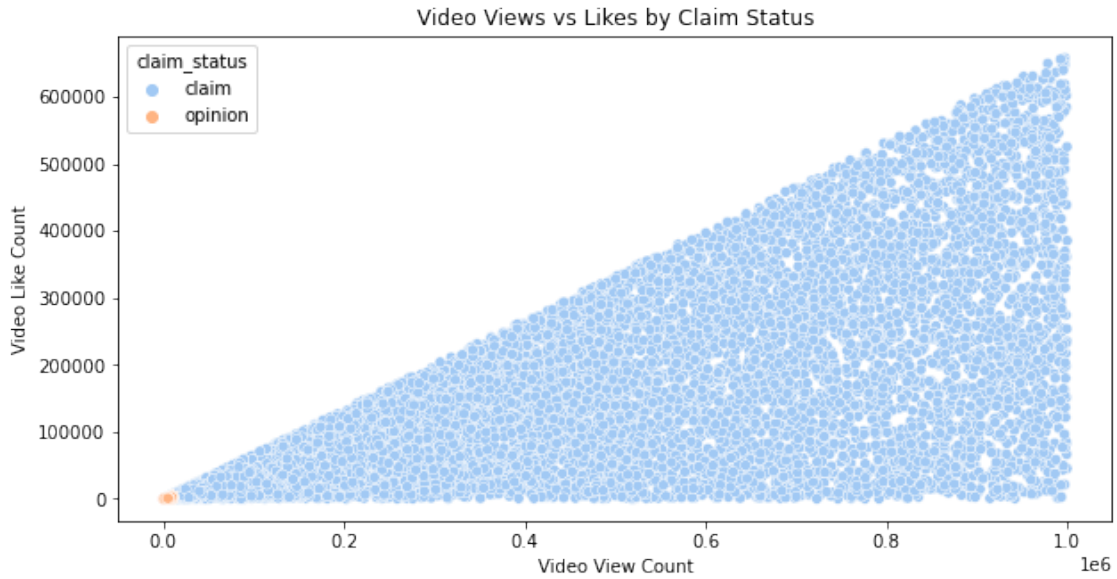
```
Example:
Number of outliers, video_view_count: ___
Number of outliers, video_like_count: ___
Number of outliers, video_share_count: ___
Number of outliers, video_download_count: ___
Number of outliers, video_comment_count: ___
```

[59]:
```python
# Define the count columns
count_cols = ['video_view_count',
                 'video_like_count',
                 'video_share_count',
                 'video_download_count',
                 'video_comment_count'
                ]

# Loop through each count column
for column in count_cols:
    q1 = data[column].quantile(0.25)
    q3 = data[column].quantile(0.75)
    iqr = q3 - q1
    median = data[column].median()
    outlier_threshold = median + 1.5*iqr

# Count the number of values that exceed the outlier threshold
    outliers_count = (data[column] > outlier_threshold).sum()
    print(f'Number of outliers, {column}:', outliers_count)
```

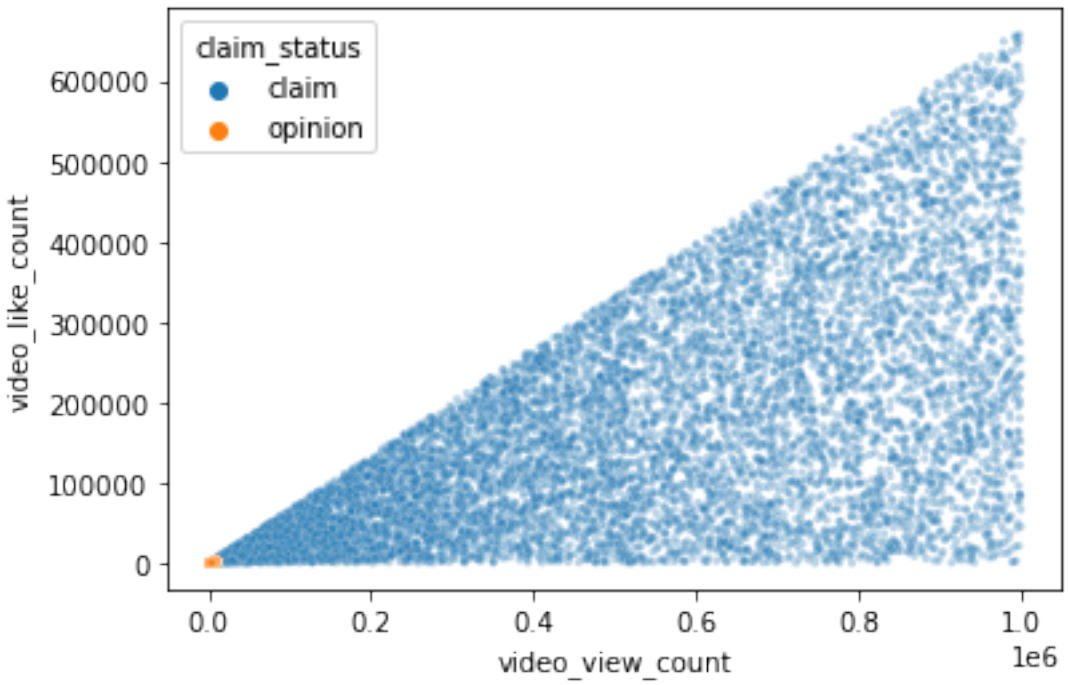
```
Number of outliers, video_view_count: 2343
Number of outliers, video_like_count: 3468
Number of outliers, video_share_count: 3732
Number of outliers, video_download_count: 3733
Number of outliers, video_comment_count: 3882
```

Scatterplot

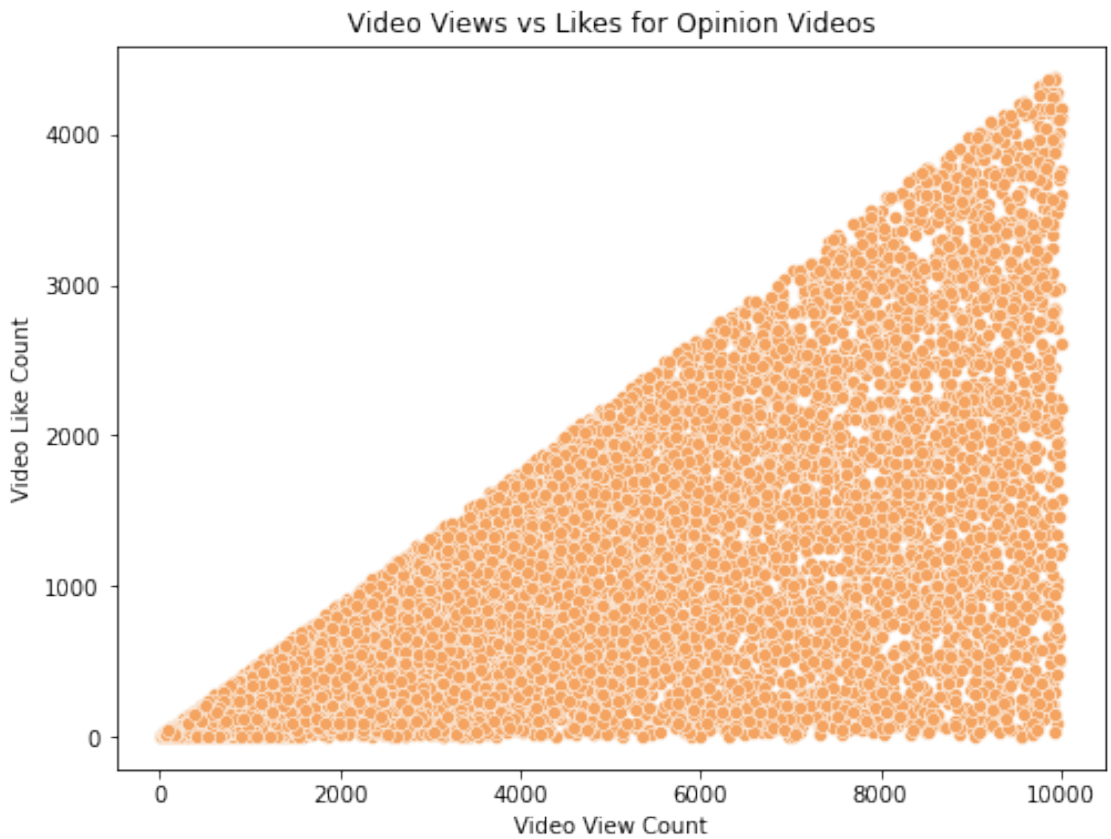
[27]:
```python
# Create a scatterplot of `video_view_count` versus `video_like_count`␣
↪according to 'claim_status'
plt.figure(figsize=(10,5))
sns.scatterplot(data=data, x='video_view_count', y='video_like_count',␣
↪hue='claim_status', palette='pastel')
plt.title('Video Views vs Likes by Claim Status')
plt.xlabel('Video View Count')
plt.ylabel('Video Like Count');
```



[62]:
```
# Create a scatterplot of `video_view_count` versus `video_like_count`␣
↪according to 'claim_status' (Exemplar)

"""
Note:
This scatterplot visualizes the relationship between 'video_view_count'
and 'video_like_count', colored by 'claim_status'.

Unlike the previous version, this example customizes the point style for
large datasets using 's' and 'alpha' parameters:

- s=10 → sets smaller point size to reduce overlap between thousands of data␣
↪points.
- alpha=0.3 → adds transparency, allowing dense regions to appear darker
  and helping reveal patterns of concentration.

Together, these settings improve clarity and make correlation patterns
between views and likes easier to observe, even with massive datasets.
"""

sns.scatterplot(x=data["video_view_count"], y=data["video_like_count"],
                hue=data["claim_status"], s=10, alpha=.3);
```

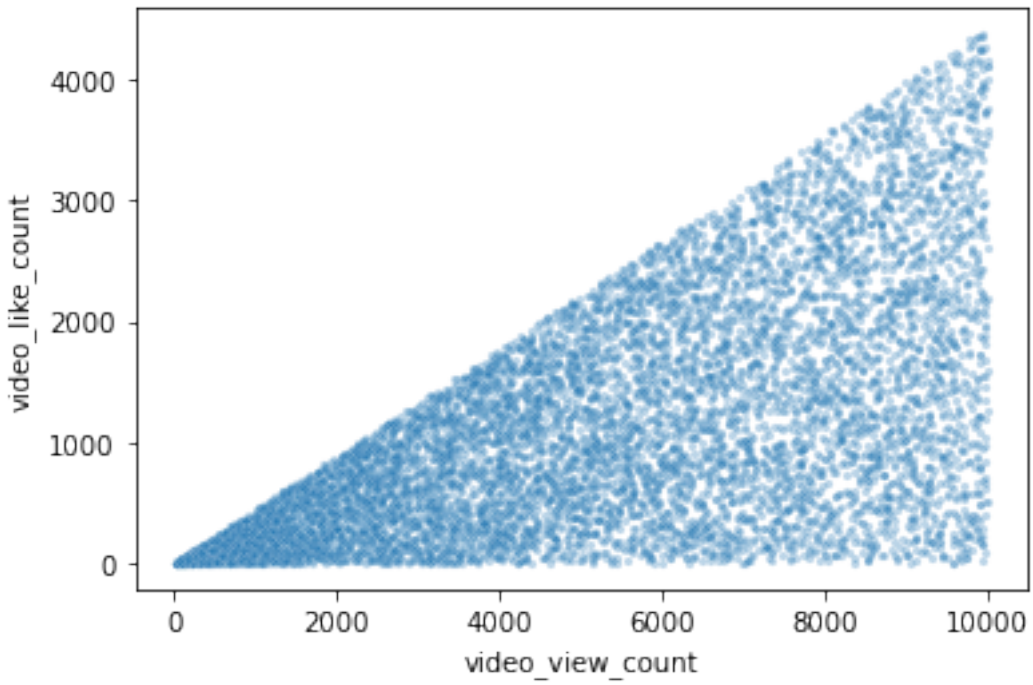


[28]:
```
# Create a scatterplot of ``video_view_count` versus `video_like_count` for␣
↪opinions only
plt.figure(figsize=(8,6))
sns.scatterplot(data=data[data['claim_status'] == 'opinion'],␣
↪x='video_view_count', y='video_like_count', color='sandybrown')
plt.title('Video Views vs Likes for Opinion Videos')
plt.xlabel('Video View Count')
plt.ylabel('Video Like Count');
```



[63]:
```
# Create a scatterplot of `video_view_count` versus `video_like_count` for␣
↪opinions only (Exemplar)
opinion = data[data['claim_status']=='opinion']
sns.scatterplot(x=opinion["video_view_count"], y=opinion["video_like_count"],
                s=10, alpha=.3)
```

[63]: <matplotlib.axes._subplots.AxesSubplot at 0x7ade6a91d050>



You can do a scatterplot in Tableau Public as well, which can be easier to manipulate and present. If you'd like step by step instructions, you can review the instructions linked in the previous Activity page.

4.4 PACE: Execute

Consider the questions in your PACE Strategy Document to reflect on the Execute stage.

4.4.1 Task 5a. Results and evaluation

Having built visualizations in Tableau and in Python, what have you learned about the dataset? What other questions have your visualizations uncovered that you should pursue?

Pro tip: Put yourself in your client's perspective, what would they want to know?

Use the following code cells to pursue any additional EDA. Also use the space to make sure your visualizations are clean, easily understandable, and accessible.

Ask yourself: Did you consider color, contrast, emphasis, and labeling?

I have learned that claim videos generally receive much higher engagement than opinion videos, especially in terms of views and likes. The data also shows a strong positive correlation between views and likes.

My other questions are about what factors might influence a video's engagement beyond claim status, such as author verification or ban status.

My client would likely want to know what characteristics make claim videos more engaging and how this insight can be used to improve content strategy.

4.4.2 Task 5b. Conclusion

Make it professional and presentable

You have visualized the data you need to share with the director now. Remember, the goal of a data visualization is for an audience member to glean the information on the chart in mere seconds.

Questions to ask yourself for reflection: Why is it important to conduct Exploratory Data Analysis? What other visuals could you create?

EDA is important because it allows analysts to understand the structure, quality, and key patterns of the data before modeling. It helps identify relationships, detect outliers, and uncover potential data quality issues. This step ensures that decisions and predictive models are built on accurate and well-understood information.

Visualizations helped me understand how engagement metrics such as views, likes, and shares differ between claim and opinion videos. They revealed that claim videos tend to attract significantly more audience interaction. Through visual exploration, I gained clear insights into which variables are most relevant for predicting claim status and which data points might skew the analysis.

You've now completed a professional data visualization according to a business need. Well done! Be sure to save your work as a reference for later work in Tableau.

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.